

MambaNA: A state space model with Neighborhood Attention for Medical Super-Resolution

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Abstract

*In recent years, due to Mamba’s ability to balance global receptive fields and computational efficiency, Mamba-based methods have demonstrated significant potential in various tasks, including medical image super-resolution. However, the inherent causal modeling of Mamba restricts its local feature extraction, this poses a challenge for medical image super-resolution which often involves intricate local details in structures like cells and tissue boundaries. In this work, we propose MambaNA, a novel medical image super-resolution method that integrates neighborhood attention mechanism into Mamba model. Specifically, MambaNA computes pixel-wise attention weights for neighboring elements to introduce non-causality, while extracting local features that maintain translational equivariance. Furthermore, it performs a multi frequency channel fusion directly following SSM’s sequence modeling, which significantly boosts model performance and training stability. Extensive experiments demonstrated the superior performance of MambaNA in medical image super-resolution tasks. **Keywords:** Medical imaging · Super-Resolution · Mamba*

1. Introduction

Modern medical diagnosis heavily relies on the visual information provided by medical imaging. However, obtaining high-quality images often involves costly advanced equipment and extended patient exposure. Deep learning advances have made super-resolution (SR) an economically efficient solution for image enhancement, capable of intelligently reconstructing fine structures from low-resolution inputs. In the medical domain, SR specifically focuses on generating high-resolution outputs from low-resolution scans by inferring and synthesizing intricate details typically lost during acquisition.

Early super-resolution research, relying on simple interpolation [10, 12, 25] and optimization [48] methods, was straightforward but often failed to precisely restore im-

age details. Subsequently, Convolutional Neural Networks (CNNs) [7, 8, 27, 29, 63, 64] and Generative Adversarial Networks (GANs) [27, 52, 53] gained widespread application due to their ability to learn nonlinear mapping relationships and representational features. However, these methods frequently struggled with capturing long-range dependencies and global receptive fields, which limited their ability to restore high-frequency details and maintain structural coherence, leading to sub-optimal reconstruction quality, especially in complex medical imagery.

To mitigate these limitations, the Transformer architecture [50], with its powerful self-attention mechanism for global dependency modeling, has been successfully adapted to computer vision, notably through the Vision Transformer (ViT) [2, 6, 9, 30, 54], demonstrating efficacy in capturing long-range spatial relationships. Despite this strength, Transformer-based methods often incur considerable computational and memory costs, largely due to the quadratic scaling of their global self-attention. State Space Models (SSMs) [16, 17, 47] have emerged as a class of deep sequence models designed to efficiently process long sequences by transforming inputs into a hidden state space. Among these, the Mamba architecture [15] stands out as a notable advancement, effectively balancing global receptive fields with computational efficiency. However, Mamba’s inherent causal modeling limits its ability to fully exploit vital local features [19], a constraint especially pronounced in medical images where fine, localized details are crucial for precise analysis.

To address these crucial limitations, we propose MambaNA, a novel medical image super-resolution method. It integrates Neighborhood Attention (NA) [21] into a Mamba architecture to introduce non-causal modeling and mitigate Mamba’s inherent causal limitations, while concurrently enhancing its local feature extraction. Furthermore, MambaNA incorporates a lightweight multi-spectral channel attention (MSCA) [40] based on the Discrete Cosine Transform (DCT), which extends traditional channel attention by fusing multi-frequency components. This allows the model to capture intricate frequency-domain patterns and cross-

channel dependencies, and experimental results show that this incorporation significantly boosts MambaNA's performance and training stability. The main contributions of this paper are summarized as follows:

1. We build a Mamba-based network for medical image super-resolution called MambaNA. It combines the advantages of Mamba's global receptive fields and spatial attention mechanism's non-causal modeling, achieving greater efficiency and accuracy in processing complex medical images.
2. We propose an SSM-Layer which incorporates a channel attention mechanism that directly follows the SSM module. It captures intricate patterns and suppresses noise by adaptively weighting different frequency components within channels, thereby boosting MambaNA's overall performance and enhancing its training stability.
3. We conducted extensive experiments on various tasks and performed ablation studies to verify each key component's contribution. The results show that MambaNA outperforms existing state-of-the-art methods across several key metrics, demonstrating its effectiveness and robustness across different scenarios.

2. Related Works

2.1. Medical Image Super-Resolution

While traditional methods often lack detail fidelity and artifact suppression, deep learning has profoundly advanced medical image SR, yielding numerous CNN-based and GAN-based algorithms.

CNNs are widely adopted for medical image SR due to their powerful feature extraction and end-to-end learning. Early works like SRCNN [8] and VDSR [26] were extended to medical contexts such as CT and MRI [38, 49]. Subsequent advancements incorporated sophisticated architectural designs. These include various forms of residual learning [37, 39, 41, 45, 65] and dense connections [3, 5, 42]. Such innovations facilitate deeper network training and enhance feature reuse, significantly improving performance across diverse medical image modalities for clinical applications [20, 44]. GANs excel at generating perceptually richer HR medical images and can mitigate the over-smoothing seen with CNNs. For instance, GANs have been utilized for denoising and SR of low-dose CT [56, 60], or combined with CycleGAN for improved CT reconstruction [61]. Additionally, architectural integrations like DCSRNet within the GAN framework [4], and specific adversarial networks for CMRI [35] or compressed sensing MRI [59], demonstrate enhanced performance.

Transformer architectures have recently gained attention in SR for their ability to model long-range dependencies and global semantic features better than traditional CNNs. For instance, models like SuperFormer [11]

and TransMRSR [23] explored combining CNNs' local strengths with Transformer-based global modeling for 3D MRI. Others such as [67] and [28] introduced frequency-based strategies for MRI, while [1] leveraged Vision Transformers in self-supervised microscopy SR. However, most of these methods require substantial GPU memory and training cost. Despite this, their ability to model global context makes them valuable for SR tasks that demand high structural fidelity.

2.2. State Space Models

SSMs have recently attracted increasing interest for sequence modeling due to their potential to overcome the efficiency limitations of RNNs and Transformers on long sequences. The Mamba architecture [15] introduces a Selective State Space Model (Selective SSM) that adaptively modulates state space parameters based on input content, enabling efficient long-range dependency modeling. It has demonstrated competitive or even superior performance to Transformers in language modeling tasks [57].

Despite Mamba's strong sequence modeling performance, its direct application to vision tasks faces challenges due to the 2D non-causal nature of image pixels and its inherent causal modeling. This causality limits full pixel utilization and can lead to decayed long-range dependencies. Researchers have addressed this by developing various scanning strategies to adapt 2D image data for Mamba's 1D causal processing. For instance, VMamba [33] pioneered cross-scan strategies for general vision tasks. ViM [68] employs bidirectional scanning with position embeddings, while PlainMamba [58] utilizes continuous 2D scanning and direction-aware updates to capture spatial relationships. Furthermore, LocalMamba [24] introduces dynamic, layer-specific optimal scan choices within distinct image windows. Building upon these advancements, Mamba-based models have rapidly become prominent in various computer vision domains [18, 19, 43, 55, 66]. These innovations enable SSMs to effectively process 2D visual data, capture global context, and achieve competitive performance against Transformers [57, 62].

2.3. Attention Mechanisms

In deep learning models, channel attention mechanisms enable dynamic focus on critical input features. They model inter-channel relationships, learning adaptive weights to efficiently allocate computational resources and enhance feature representation. SENet [22] pioneered this field with its Squeeze-and-Excitation (SE) module, which aggregates global spatial information via global average pooling and then learns channel dependencies. To mitigate information loss from traditional global average pooling, FcaNet [40] introduced Multi-Spectral Channel Attention using Discrete Cosine Transform for frequency-domain analysis. Further-

more, ECA-Net [51] employs a 1D convolution for efficient local cross-channel interaction without dimensionality reduction.

Spatial attention mechanisms, on the other hand, highlight the importance of different spatial locations within a feature map. The emergence of ViT [9] marked a paradigm shift in computer vision, demonstrating that pure self-attention architectures could achieve competitive performance with CNNs by modeling long-range dependencies. Swin Transformer [34] introduced a Shifted Window (Swin) Attention, which computes self-attention within non-overlapping local windows and allows for cross-window connections through a shifted window partitioning approach. Beyond these general advancements, the latest efforts, highlight a strong emphasis on synergistic and dual attention mechanisms to enhance feature representation across various vision tasks, as exemplified by works such as SCSA [46], DACN [36], CATA-TCN [32], and TSCA-Net [13], demonstrating powerful potential in areas like super-resolution and medical image processing.

3. Methodology

3.1. Preliminaries

3.1.1. State Space Models

SSMs are a class of models inspired by control systems theory for sequence modeling. They have recently gained attention in deep learning for their ability to handle long-range dependencies efficiently. The core idea of SSMs is to model the relationship between an input sequence $x \in \mathbb{R}$ and an output sequence $y \in \mathbb{R}$ through a hidden state $h \in \mathbb{R}^N$. Formally, a continuous-time SSM can be defined by a set of linear ordinary differential equations (ODEs):

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t), \\ y(t) &= Ch(t) + Dx(t), \end{aligned} \quad (1)$$

where N is the state size, $A \in \mathbb{R}^{N \times N}$, $B \in \mathbb{R}^{N \times 1}$, and $C \in \mathbb{R}^{1 \times N}$ are the system matrices governing the state dynamics and output mapping, and $D \in \mathbb{R}$ is the skip connection weight.

To make SSMs applicable in deep learning scenarios, the continuous-time model is discretized. Using the zero-order hold (ZOH) assumption, the continuous-time parameters A, B are converted to discrete parameters $\bar{A}, \bar{B}$ over a specified sampling interval $\Delta \in \mathbb{R} > 0$:

$$\begin{aligned} \bar{A} &= e^{\Delta A}, \\ \bar{B} &= (\Delta A)^{-1}(e^{\Delta A} - I) \cdot \Delta B. \end{aligned} \quad (2)$$

This leads to a discrete-time model formulation:

$$\begin{aligned} h_t &= \bar{A}h_{t-1} + \bar{B}x_t, \\ y_t &= Ch_t + Dx_t. \end{aligned} \quad (3)$$

The discretized form allows SSMs to be integrated into deep learning architectures efficiently, and Eq. (3) can be accelerated via parallel computation, leveraging a global convolution operation:

$$\begin{aligned} \bar{K} &= (C\bar{B}, C\bar{A}\bar{B}, \dots, C\bar{A}^{L-1}\bar{B}), \\ y &= x \circledast \bar{K}, \end{aligned} \quad (4)$$

where $\circledast$ stands for the convolution operation, L is input sequence's length, and $\bar{K} \in \mathbb{R}^L$ represents the convolution kernel. By utilizing convolution, this method enables the simultaneous computation of outputs across the sequence, which significantly improves computational efficiency and scalability.

3.1.2. Mamba: Selective State Space Models

Mamba [15] is a novel variant of SSMs that extends the S4 framework [16] by incorporating a data-dependent parameterization. This allows Mamba to leverage the linear computational complexity with respect to the input sequence length for highly efficient, content-aware processing of long sequences. Mamba surpasses Transformers in performance on large-scale real data, showcasing its effectiveness across various sizes and demonstrating linear scalability in sequence length.

The key innovation of Mamba lies in its selective mechanism, which adjusts the parameters B, C, Δ based on the input sequence. This allows the model to dynamically capture the characteristics of long-sequence signals using a simple architecture. The selective mechanism uses the input $x \in \mathbb{R}^{B \times L \times D}$ to generate input-dependent parameters:

$$\begin{aligned} B &\in \mathbb{R}^{B \times L \times N} \leftarrow s_B(x), \\ C &\in \mathbb{R}^{B \times L \times N} \leftarrow s_C(x), \\ \Delta &\in \mathbb{R}^{B \times L \times D} \leftarrow s_\Delta(x), \end{aligned} \quad (5)$$

where $s(\cdot)$ is typically implemented using a linear projection layer. Mamba shares the same recursive form of the hidden state update Eq. (3), which enables the model to memorize ultra-long sequences so that more pixels can be activated to aid restoration.

For exceptional efficiency, Mamba introduces a specialized parallel algorithm for its recurrent mode, this algorithm is meticulously designed to optimize memory access and minimize redundant computations on modern accelerators, thereby ensuring efficient inference. Complementing this, Mamba adopts a simplified end-to-end neural network architecture that entirely eschews attention modules and even traditional MLP blocks. This architectural parsimony, centered around the selective SSM layer, significantly reduces parameter count and computational overhead, leading to faster training and inference times.

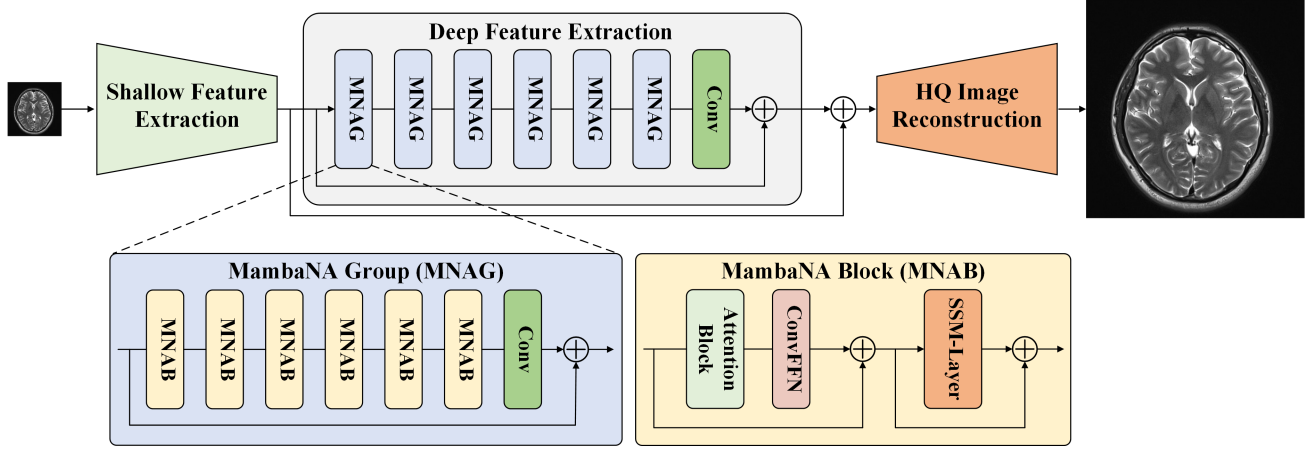


Figure 1. Overall Architecture

3.2. Overall Architecture

As depicted in Figure 1, MambaNA is composed of three modules: shallow feature extraction, deep feature extraction and HQ image reconstruction modules, which aligns with the prevalent architectural paradigms of image restoration Transformer networks [30]. Given a LQ input image $I_{LQ} \in \mathbb{R}^{H \times W \times C_{in}}$, it first undergoes shallow feature extraction, typically implemented as a single 3×3 convolutional layer. This process transforms the input into initial features $F_0 \in \mathbb{R}^{H \times W \times C}$, which retain the original spatial dimensions while expanding the channel depth. The subsequent deep feature extraction stage, where the core processing and our novel enhancements are applied, is responsible for extracting complex hierarchical features.

Specifically, this stage comprises a stack of N_1 residual MambaNA groups (MNAGs). Within each MNAG, N_2 cascaded MambaNA Blocks (MNABs) are utilized for core feature transformation. Each MNAB, in turn, is composed of an Attention Block and a SSM-layer, collectively performing the detailed feature extraction and refinement. Upon completion of this deep feature extraction stage, the initial features F_0 are progressively transformed into refined deep features $F_{N_1} \in \mathbb{R}^{H \times W \times C}$. Finally, the HQ image reconstruction module leverages these refined deep features to progressively reconstruct the desired HQ output image.

For the task of medical image super-resolution, MambaNA is optimized using a hybrid loss function combining ℓ_1 loss and perceptual loss. The final loss is defined as:

$$\begin{aligned} \mathcal{L} &= 0.7 \cdot \mathcal{L}_{L_1} + 0.3 \cdot \mathcal{L}_{perceptual}, \\ \mathcal{L}_{L_1} &= \|I_{HQ} - I_{GT}\|_1, \\ \mathcal{L}_{perceptual} &= \|\phi(I_{HQ}) - \phi(I_{GT})\|_2^2 \end{aligned} \quad (6)$$

where $\phi(\cdot)$ denotes feature extraction from pre-trained VGG-19's 5th conv layer.

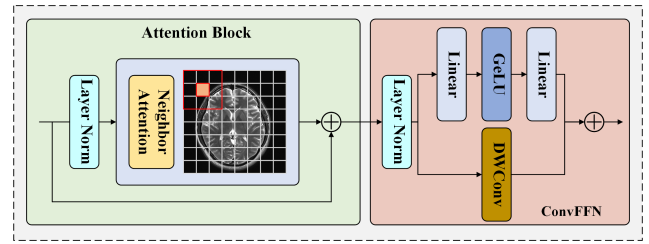


Figure 2. MambaNA's Attention Block

3.3. Attention Block

MambaNA incorporates the Neighborhood Attention (NA) mechanism [21] as a pivotal attention extraction module to incorporate non-causal modeling ability and at the same time, enhance the perception of intricate image details.

NA attention employs a sliding window paradigm to perform self-attention operations. Unlike global self-attention or conventional window-based attention, NA strictly confines each query pixel's attention computation to its predefined fixed-size local neighborhood. Specifically, given an input $X \in \mathbb{R}^{N \times d}$, it will continuously calculate the adjacent attention of the i -th input:

$$\begin{aligned} A_i^k &= Q_i K_{\mathcal{N}(i)}^T + B_i, \\ NA_k(i) &= \text{softmax}\left(\frac{A_i^k}{\sqrt{d}}\right) V_{\mathcal{N}(i)}. \end{aligned} \quad (7)$$

Here, k is the neighborhood size, $B_i \in \mathbb{R}^{1 \times k}$ is the relative positional biases, and $Q_i \in \mathbb{R}^{1 \times d}$, $K_{\mathcal{N}(i)} \in \mathbb{R}^{k \times d}$, $V_{\mathcal{N}(i)} \in \mathbb{R}^{k \times d}$ are respectively the i -th input's query, key, and value matrix where $\mathcal{N}(i)$ represents the i -th element's neighborhood. This localized design of NA inherently incorporates local inductive bias and ensures translational equivariance to patterns appearing at different positions within the image.

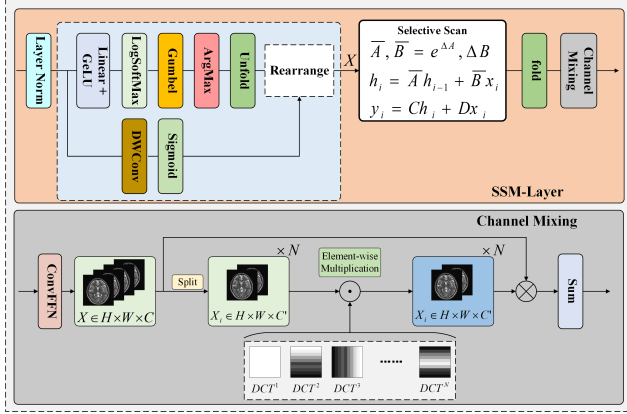


Figure 3. MambaNA's SSM-Layer

Compared to window-based attention mechanisms like Swin, NA is better suited for capturing the rich and fine local details prevalent in medical images, such as subtle vascular textures or lesion boundaries. This is primarily because Swin's fixed window partitioning can lead to high-frequency information loss or processing discontinuities at window boundaries, whereas NA's overlapping sliding windows provide a smoother, more continuous context aggregation, and its inherent translational equivariance is crucial for medical image super-resolution. Furthermore, NA also efficiently expands the receptive field through multi-layer stacking without incurring the high computational cost of global attention, making it particularly effective for high-resolution medical image processing without complex shifting operations as seen in Swin.

Theoretically, the integration of NA attention with Mamba models can yield a powerful synergy. Mamba models excel at efficiently capturing long-range dependencies and global structural information through their selective scan mechanisms, thereby ensuring overall image coherence. However, being fundamentally 1D sequence-based, their ability to model dense, non-linear local spatial relationships in 2D images might be less specialized than dedicated local operations. NA attention precisely complements this, focusing on providing fine-grained local inductive bias and efficient pixel-level spatial interactions. Consequently, this integration allows MambaNA to possess both remarkable global consistency and precise local detail rendering capabilities. This "global-local" synergistic processing paradigm is especially vital for medical image super-resolution, as it necessitates both the restoration of large-scale anatomical integrity and the meticulous reconstruction of minute pathological details to ensure diagnostic accuracy.

3.4. SSM-Layer

In MambaNA, the SSM-Layer is designed to efficiently capture long-range dependencies with the state space equation. Conventional selective SSMs often process 2D image data by linearizing it into 1D sequences via fixed, data-independent scanning sequences. This rigid methodology often hinders a query pixel from effectively utilizing spatially distant yet semantically similar already-processed pixels, thereby failing to capture comprehensive non-causal global context. To address this limitation and circumvent the inherent inefficiencies and computational redundancy associated with employing multiple fixed scanning sequences, SSM-Layer employs a dynamic pixel scanning strategy predicated on semantic categorization.

Given feature map $X \in \mathbb{R}^{C \times H \times W}$, we first compute a semantic relevance score for each pixel:

$$s(i) = MLP(GAP(X_i)), \quad (8)$$

where i indexes spatial locations and GAP extracts local statistics from a $k \times k$ neighborhood. Pixels are then softly grouped by quantizing $s(i)$ into G semantic bins:

$$g(i) = \lfloor G \cdot \sigma(s(i)) \rfloor. \quad (9)$$

For each group, pixels preserve their 2D raster order, while the final 1D scan sequence π is formed by concatenating groups from low to high semantic relevance:

$$\pi = \text{concat}(\pi_{g=0}, \pi_{g=1}, \dots, \pi_{g=G-1}). \quad (10)$$

This adaptive ordering reduces semantic distance in the SSM sequence while maintaining $O(HW)$ complexity.

After assigning a definitive semantic category to each pixel, these pixels are then reordered according to their respective categories to construct the final one-dimensional scanning sequence. This methodology ensures that semantically similar pixels are prioritized during the state update process within the SSM, thereby mitigating the weakening effect of remote pixel interactions caused by long-range decay.

Given that medical images encompass rich details, textures, and overall structures, which correspond to high, mid, and low-frequency information respectively, we strategically integrated the Multi-Spectral Channel Attention (MSCA) module subsequent to the SSM-Layer. Specifically, MSCA precisely extracts multiple frequency components from the spatial information within each channel by applying a fixed bank of pre-computed 2D DCT basis functions as filters. These extracted frequency responses then form the multi-spectral content, based on which MSCA learns scaling factors w for channels:

$$\begin{aligned} \text{Freq}^i &= \sum_{h=0}^{H-1} \sum_{w=0}^{W-1} X^{i,h,w} \cdot DCT^{i,h,w}, \\ \text{Freq} &= \text{cat}([\text{Freq}^1 \quad \text{Freq}^2 \quad \dots \quad \text{Freq}^C]), \end{aligned} \quad (11)$$

$$w = \sigma(MLP(Freq)). \quad (12)$$

Here, $X \in \mathbb{R}^{C \times H \times W}$ is the input feature map, and $DCT \in \mathbb{R}^{C \times H \times W}$ is the 2D DCT basis filter.

The integration of the MSCA module significantly boosts MambaNA’s performance and its training stability. This stability enhancement fundamentally arises from MSCA’s capacity for frequency-aware feature conditioning and adaptive channel regulation, which is particularly pertinent for SSMs. While SSM adeptly aggregates global spatio-sequential information, its inherent one-dimensional processing of two-dimensional image data can render it acutely sensitive to the conditioning, noise, and distributional properties of individual feature channels, potentially propagating instabilities through recurrent state updates. By instilling robust channel-wise feature representations, MSCA effectively acts as a frequency-domain filter and provide a powerful adaptive normalization and magnitude control for each channel, enabling optimizers to converge more reliably and efficiently.

4. Experiments

4.1. Experimental Settings

4.1.1. Datasets and Metrics

Our experiments utilized two publicly available medical image super-resolution datasets: IXI¹ dataset and a dedicated Brain Tumor MRI dataset². The IXI dataset includes 3D multimodal MRI scans (T1, T2, PD modalities). We focused on the T1-weighted subset (IXI-T1) for SR training/testing, given its superior brain soft tissue contrast. Each original IXI-T1 scan contains 96 slices (240×240 pixels); we uniformly selected 8 key slices per scan to avoid redundant low-variation samples. The IXI-T1 dataset was split as follows: 600 slices for training, 64 slices for validation, and 84 slices from remaining qualified T1 scans for testing. All slices retain the 240×240 pixel resolution.

To quantitatively evaluate the quality of the super-resolved images, we selected the widely recognized metrics: Peak Signal-to-Noise Ratio (PSNR) and the Structural Similarity Index Measure (SSIM). PSNR measures pixel-wise fidelity by comparing reconstructed images to ground truth, while SSIM assesses perceptual similarity based on luminance, contrast, and structure.

4.1.2. Experimental Details

Our MambaNA is implemented with the PyTorch toolbox and trained on NVIDIA RTX A6000 GPU. The training low-resolution samples are generated by the frequency domain method, ensuring a closer alignment with the actual distribution of low-resolution images. We adopt Adam as

the optimizer with $\beta_1 = 0.9, \beta_2 = 0.99$. The initial learning rate and batch size are respectively set at 2×10^{-4} and 2. Additionally, we crop the original images into 64×64 patches during training. We set up six Mamba blocks and each block’s channel count is 192.

4.2. Comparison on Image Super-Resolution

Our comprehensive evaluation first focused on validating MambaNA’s performance against existing state-of-the-art (SOTA) medical image super-resolution methods on the challenging IXI dataset. As detailed in Table 1, MambaNA consistently achieves superior PSNR and SSIM scores compared to [14], [19], and [30]. These quantitative results underscore MambaNA’s exceptional ability to reconstruct high-fidelity medical images, effectively balancing global context modeling with precise local detail recovery.

Beyond numerical metrics, Figure 4 provides a comprehensive qualitative comparison of the super-resolution results: the first row displays the reconstructed images from various methods, while the second row presents pixel-wise mean squared error (MSE) maps between each super-resolved image and the ground truth, allowing for an intuitive understanding of reconstruction accuracy. Darker regions in the error maps indicate smaller deviations from the ground truth, signifying better reconstruction fidelity. As visually demonstrated, MambaNA produces reconstructions notably closer to the ground truth, with error maps revealing a significant reduction in artifacts and a more accurate preservation of finer anatomical structures compared to other methods that often exhibit blurring or false details in challenging areas. This pronounced qualitative superiority further reinforces MambaNA’s practical utility in clinical image analysis.

To assess MambaNA’s generalization capabilities, we also evaluated its performance on widely recognized natural image super-resolution benchmark datasets, including BSDS100, Urban100, and Manga109. As presented in Table 3, MambaNA consistently achieves state-of-the-art (SOTA) or near-SOTA performance in terms of PSNR and SSIM across these datasets, demonstrating its competitiveness against various established natural image super-resolution algorithms. This indicates that MambaNA’s design, while optimized for medical images, effectively transfers its robust super-resolution capabilities to diverse natural image scenarios, highlighting its versatility and broad applicability.

4.3. Ablation Study

We conducted a comprehensive series of ablation experiments, aiming to validate the effectiveness of the Attention Block module and Channel Mixing module in enhancing super-resolution performance and model stability.

Our experimental setup involved a precise three-step ab-

¹<https://brain-development.org/ixi-dataset/>

²<https://www.kaggle.com/datasets/masoudnickparvar/brain-tumor-mri-dataset>

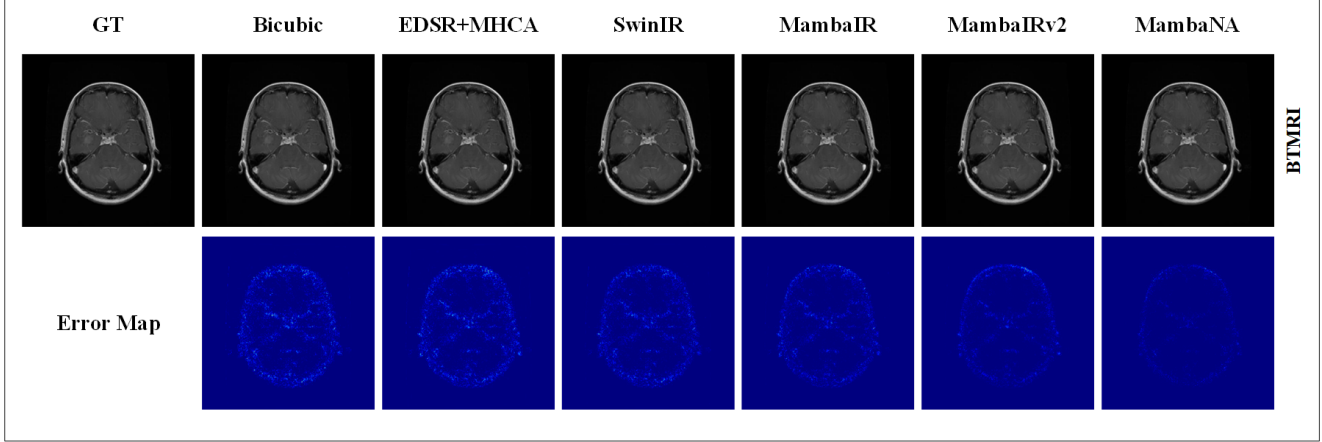


Figure 4. Error Map

Table 1. Comparison on Medical Image Super-Resolution.

Method	Scale	Param	Brain Tumor		IXI PD	
			PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
Bicubic [2D]	2 $\times$	-	26.96	0.9331	27.57	0.9523
EDSR [31] + MMHCA [14]	2 $\times$	20.8M	31.70	0.8397	32.08	0.9604
SwinIR [30]	2 $\times$	11.9M	35.60	0.9347	32.82	0.9631
MambaIR [18]	2 $\times$	22.9M	35.75	0.9362	33.30	0.9650
MambaIRv2 [19]	2 $\times$	27.6M	36.43	0.9431	33.58	0.9664
MambaNA [Ours]	2 $\times$	30.5M	36.95	0.9447	33.94	0.9686
Bicubic [2D]	4 $\times$	-	18.85	0.7535	21.22	0.8293
EDSR [31] + MMHCA [14]	4 $\times$	20.8M	28.10	0.9059	28.45	0.8925
SwinIR [30]	4 $\times$	11.9M	30.23	0.9235	28.12	0.9011
MambaIR [18]	4 $\times$	22.9M	30.75	0.9202	29.56	0.9073
MambaIRv2 [19]	4 $\times$	27.6M	31.50	0.8609	30.07	0.9334
MambaNA [Ours]	4 $\times$	30.5M	32.28	0.9386	30.64	0.9379

Attention Block	Channel Mixing	PSNR $\uparrow$	SSIM $\uparrow$
$\times$	$\checkmark$	36.41	0.9413
$\checkmark$	$\times$	36.65	0.9462
$\times$	$\times$	35.68	0.8874

Table 2. Ablation on the effectiveness of different components.

lation process. For the first step, the Attention Block module was directly removed, allowing us to ascertain the impact of non-causal modeling and enhanced local feature extraction. In the second step, we removed the Channel Mixing module in SSM-Layer, to understand its specific contribution to channel-wise feature refinement and overall model robustness. Finally, the third step involved testing a baseline model, where both Attention Block and Channel Mix-

ing modules were simultaneously removed, representing a simplified Mamba-based super-resolution architecture. All ablation variants were trained and evaluated under identical conditions, including dataset splits, optimizer, learning rate schedule, and epoch count, ensuring a fair and rigorous comparison. Performance was quantified using PSNR and SSIM.

The results, as summarized in Table 2, clearly underscore the critical and independent contribution of both the Attention Block and the Channel Mixing module. Removing the Attention Block led to a notable performance degradation of 1.0dB in PSNR compared to the full MambaNA model. This significant drop directly validates Attention Block’s pivotal role in introducing non-causal modeling and enhancing local feature extraction. Similarly, ablating the Channel Mixing module also resulted in a performance decrease of 0.8dB in PSNR. This confirms Channel Mixing’s

Table 3. Comparison on Classical Image Super-Resolution.

Method	Scale	Param	BSDS100		Urban100		Manga109	
			PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$	PSNR $\uparrow$	SSIM $\uparrow$
EDSR [31]	2 $\times$	42.6M	32.32	0.9013	32.93	0.9351	39.10	0.9773
SwinIR [30]	2 $\times$	11.8M	31.63	0.9022	32.98	0.9363	38.89	0.9736
MambaIR[18]	2 $\times$	20.4M	31.57	0.9302	34.06	0.9446	40.10	0.9784
MambaIRv2 [19]	2 $\times$	22.9M	32.71	0.9046	34.88	0.9471	40.37	0.9785
MambaNA [Ours]	2 $\times$	31.5M	32.82	0.9054	35.36	0.9493	40.81	0.9871
EDSR [31]	4 $\times$	43.0M	27.68	0.7398	26.63	0.8012	30.89	0.9011
SwinIR [30]	4 $\times$	11.9M	27.78	0.7482	27.39	0.8213	31.92	0.9136
MambaIR[18]	4 $\times$	20.4M	27.90	0.7489	27.51	0.8273	32.16	0.9245
MambaIRv2 [19]	4 $\times$	23.1M	27.86	0.7498	27.72	0.8335	32.34	0.9288
MambaNA [Ours]	4 $\times$	31.5M	27.93	0.7488	28.04	0.8511	32.55	0.9382

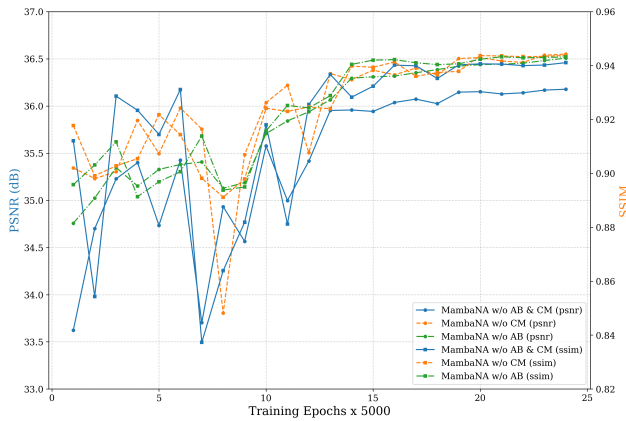


Figure 5. MambaNA Ablation Study: Training Convergence (PSNR, SSIM). This figure shows the training curves on the Brain Tumor dataset, including “MambaNA w/o CM” (without Channel Mixing), “MambaNA w/o AB” (without Attention Block) and “MambaNA w/o AB & CM” (without both), illustrating each component’s contribution to performance and stability.

effectiveness in refining channel-wise feature representations by adaptively weighting frequency components. Beyond quantitative gains, the Channel Mixing module proved indispensable for maintaining training stability, as Figure 5 vividly shows with smoother convergence and reduced oscillations. The most substantial performance decline was observed in the baseline model, which showed a 0.43dB drop in PSNR compared to the full model, highlighting the cumulative benefits of our design. These comprehensive results rigorously validate our design choices, confirming that each proposed component is vital for MambaNA’s performance and robustness in medical image super-resolution.

5. Conclusion

In this paper, we successfully developed MambaNA, a novel super-resolution framework tailored for medical image reconstruction. By strategically adopting the Mamba architecture for its inherent efficiency and long-range dependency modeling, which are critical for processing high-resolution and volumetric medical data. To overcome the inherent limitations of Mamba’s causal sequence modeling, which restricts comprehensive non-causal spatial context learning vital for precise 2D medical image detail recovery, MambaNA innovatively integrates both an Attention Block and a Channel Mixing module. The Attention Block effectively injects non-causal modeling capabilities into the Mamba backbone, and the Channel Mixing module significantly enhances the model’s training stability, these concurrently contribute to substantial improvements in overall performance. Extensive experiments consistently demonstrated MambaNA’s superior performance on medical image super-resolution tasks, achieving state-of-the-art (SOTA) or near-SOTA levels on these benchmarks. It also showed robust generalization to natural image tasks, where it also reached highly competitive performance. This work thus underscores the significant potential of Mamba-based architectures in high-fidelity image reconstruction. While promising, future work will focus on optimizing MambaNA’s computational efficiency for extremely large 3D volumes and exploring unsupervised learning strategies to reduce data dependency, thereby expanding its practical utility in diverse clinical and research environments. -

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